User Manual

Arxiv Research Paper Classifier

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1 Introduction

1.1 Overview

The **Arxiv Research Paper Classifier** is an application designed to automatically identify the subject area of **Computer Science research papers** using their title and abstract.

The system predicts categories aligned with common arXiv-style classifications such as *Machine Learning (cs.LG)*, *Computer Vision (cs.CV)*, and related domains.

Instead of manually reviewing papers and determining their area, users can obtain quick and consistent category predictions.

1.2 Why Use This Application

This tool is particularly useful in Computer Science research workflows, including:

- Organizing collections of CS research papers by domain
- Identifying the field of unfamiliar papers (e.g., ML vs Systems vs Vision)
- Assisting in dataset preparation for research or machine learning tasks
- Processing large volumes of arXiv-style metadata efficiently

1.3 Key Features

- **Single Prediction:** Classify one research paper using its title and abstract
- **Bulk Prediction:** Classify multiple papers using a CSV file
- **CS-Specific Categories:** Outputs categories aligned with Computer Science domains
- **Fast Results:** Predictions are generated within seconds
- **Simple Interface:** Designed for ease of use without technical expertise

2 Starting the Application

2.1 Step 1: Install Python

Ensure Python 3.10 is installed on your system.

2.2 Step 2: Install Dependencies

Open a terminal in the project folder and run:

```
pip install -r requirements.txt
```

2.3 Step 3: Run the Application

Start the system using:

```
python run.py
```

This command will automatically:

- Start the backend
- Launch the model server
- Start the frontend interface

2.4 Step 4: Open in Browser

Open your browser and go to:

```
http://localhost:8501
```

Figure 1: Application running in browser

2.5 Important Notes

- Do not close the terminal while using the application
- Wait until the system shows *"System Ready"*

3 Understanding the Interface

3.1 Main Layout

The application has two main parts:

- Sidebar (mode selection)
- Main panel (input and results)

3.2 Mode Selection

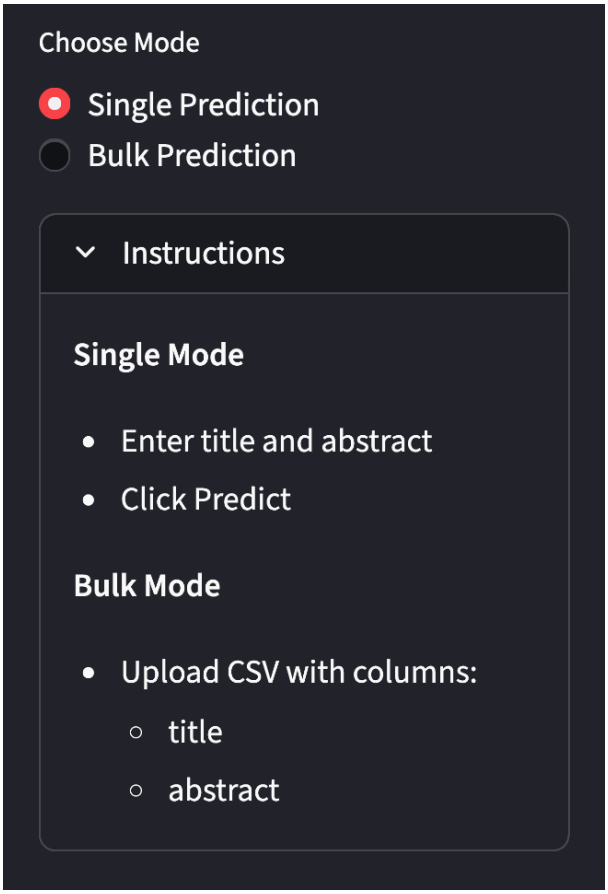

Figure 2: Selecting prediction mode

The sidebar provides two options:

- **Single Prediction**
- **Bulk Prediction**

4 Single Prediction

4.1 Overview

The **Single Prediction** mode allows users to classify an individual research paper into one or more relevant categories. This mode is ideal for quick analysis of a single title–abstract pair without requiring batch processing.

It is particularly useful for:

- Quickly identifying the domain of a research paper
- Verifying model behavior on specific examples
- Testing hypotheses or exploring edge cases

4.2 When to Use

Use this mode when:

- You have a single research paper to classify
- You want immediate feedback on a specific input
- You are experimenting with different titles/abstracts

For multiple papers, consider using the **Batch Prediction** mode instead.

4.3 Steps

Follow the steps below to generate a prediction:

1. Navigate to the **Single Prediction** tab in the application
2. Enter the **Title** of the research paper
3. Enter the **Abstract** in the input field
4. Click the **Predict** button

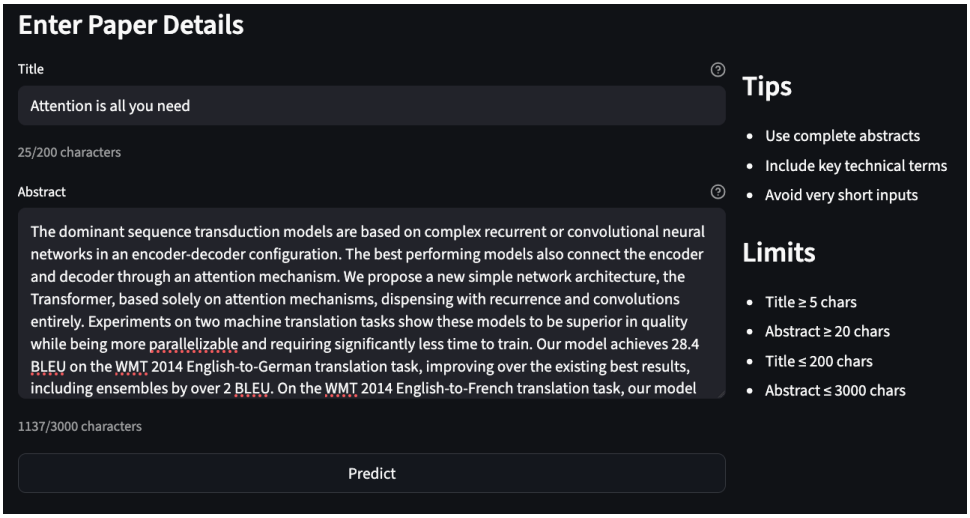

Figure 3: Input interface for entering title and abstract

4.4 Input Guidelines

To obtain accurate predictions, follow these best practices:

- **Title:** Keep it concise but informative
- **Abstract:** Provide sufficient detail about the methodology, problem, and results
- Avoid:
  - Extremely short or vague descriptions
  - Non-technical or unrelated text
  - Missing either title or abstract

4.5 Example

Title: Deep Learning for Image Classification

Abstract: This paper explores convolutional neural networks for large-scale image classification tasks. We evaluate different architectures and optimization techniques, demonstrating improved performance on benchmark datasets.

4.6 Processing

Once the **Predict** button is clicked:

- A status message such as *"Analyzing..."* is displayed
- The system preprocesses the input (tokenization, normalization, etc.)

- The trained model performs inference on the processed data

This step typically takes a few seconds depending on system load.

4.7 Results

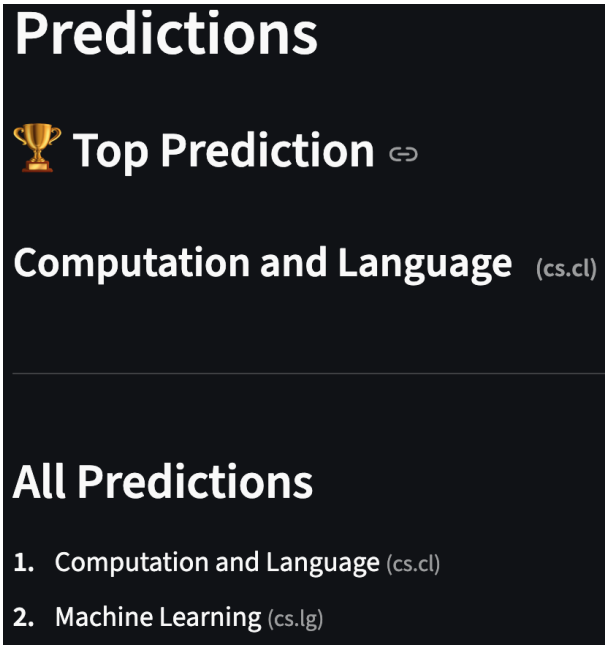

Figure 4: Predicted categories displayed to the user

The output consists of:

- One or more predicted categories (e.g., `cs.LG`, `cs.CV`)
- Categories are typically ranked based on model confidence

4.8 Interpreting Results

- The **top prediction** represents the most likely category
- Additional labels indicate possible secondary domains
- Predictions may overlap across related research areas

4.9 Common Issues and Troubleshooting

- **No predictions returned:**
  - Ensure both title and abstract are provided

- **Irrelevant predictions:**
  - Improve abstract clarity and detail
  - Avoid ambiguous terminology
- **Slow response:**
  - May occur due to system load or large input size

4.10 Tips for Best Performance

- Use domain-specific keywords (e.g., "transformer", "quantum entanglement", "reinforcement learning")
- Include methodological details when possible
- Ensure the abstract reflects the core contribution of the paper

5 Bulk Prediction

5.1 Overview

The **Bulk Prediction** mode enables users to classify multiple research papers in a single operation using a CSV file. This mode is designed for efficiency and is particularly useful when working with large datasets.

Typical use cases include:

- Annotating datasets for research or analysis
- Evaluating model performance on multiple samples
- Automating large-scale paper categorization

5.2 When to Use

Use this mode when:

- You have multiple papers to classify
- You want to process data in batches
- You are working with structured datasets (e.g., CSV files)

For single inputs, use the **Single Prediction** mode.

5.3 Preparing the CSV File

Your input file must meet the following requirements:

- File format: **CSV (.csv)**
- Required columns:
  - **title**
  - **abstract**
- Each row represents one research paper

5.4 Formatting Rules

To ensure successful processing:

- Column names must match exactly: `title`, `abstract`
- Do not leave required fields empty
- Avoid special formatting (merged cells, line breaks within fields)
- Use UTF-8 encoding to prevent character issues

5.5 Example CSV

```
title,abstract
Paper 1,This paper studies...
Paper 2,Another study...
```

5.6 Steps

Follow these steps to perform bulk prediction:

1. Navigate to the **Bulk Prediction** section
2. Upload your prepared CSV file
3. Click the **Run** (or **Predict**) button

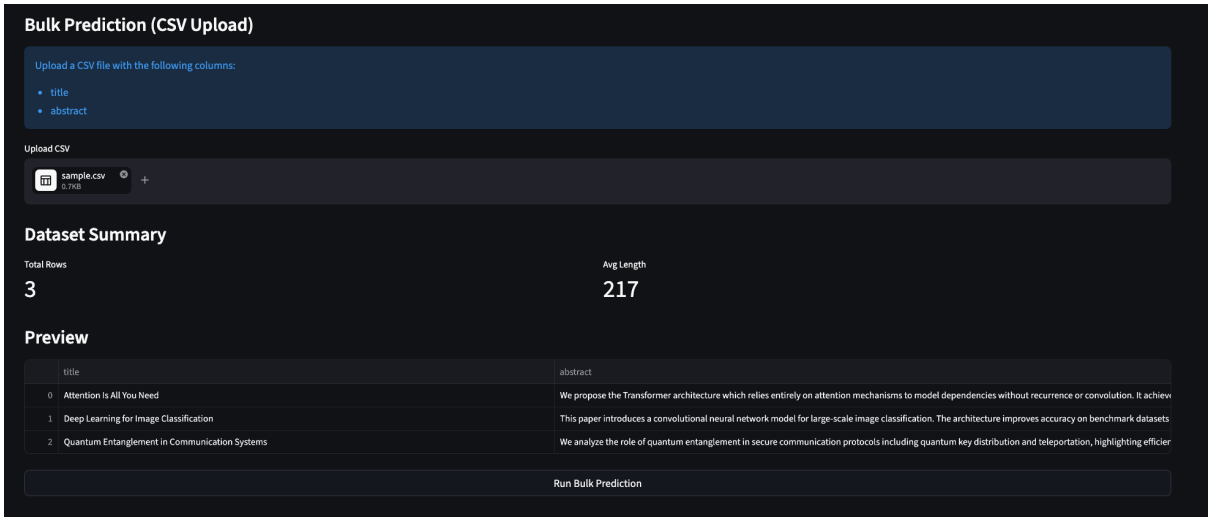

Figure 5: Uploading a CSV file for bulk prediction

5.7 Processing

Once the file is uploaded and execution begins:

- A status message such as *"Processing batch..."* is displayed
- Each row is processed sequentially or in parallel (depending on system design)
- Inputs are preprocessed (cleaning, tokenization, normalization)
- The trained model generates predictions for each entry

Processing time depends on:

- Number of rows in the CSV file
- Length of abstracts
- Available system resources

5.8 Results

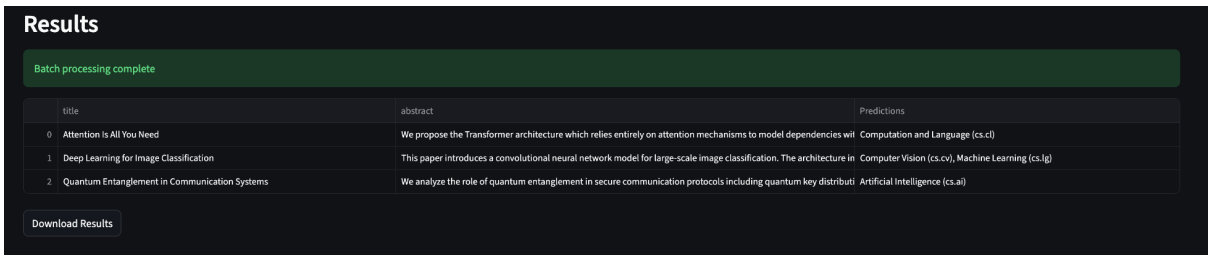

Figure 6: Predictions generated for multiple papers

The output is typically displayed as a table where:

- Each row corresponds to an input paper
- Additional columns contain predicted categories
- Predictions may include multiple labels per paper

5.9 Output Format

Depending on the system, the output may include:

- Predicted labels (e.g., `cs.LG`, `cs.AI`)
- Ranked predictions based on confidence
- Optional confidence scores (if enabled)

5.10 Downloading Results

In most implementations, you can:

- Download the results as a CSV file
- View predictions directly in the interface

5.11 Error Handling

Common issues and their solutions:

- **Missing columns:**
  - Ensure both `title` and `abstract` are present
- **Empty rows:**
  - Remove or fill incomplete entries
- **File not uploading:**
  - Check file format (.csv only)
  - Ensure file size is within limits
- **Encoding issues:**
  - Save file with UTF-8 encoding

5.12 Performance Tips

- Keep abstracts informative but concise
- Avoid extremely large files; split into smaller batches if needed
- Ensure clean, well-formatted data for best accuracy

5.13 Best Practices

- Pre-clean your dataset (remove duplicates, fix formatting)
- Validate column names before uploading
- Start with a small sample to verify correctness

6 Understanding Results

6.1 Overview

After processing, each paper is assigned one or more predicted categories representing its research domain. These categories are generated by a trained machine learning model based on the provided title and abstract.

6.2 Single vs Multiple Labels

- A **single category** indicates that the paper is strongly associated with one primary research area.
- **Multiple categories** suggest that the paper spans multiple domains or lies at the intersection of different fields (e.g., machine learning applied to biology).

6.3 Ranking and Confidence

- Predictions are typically ranked based on model confidence.
- The **top prediction** is the most likely category.
- Additional labels represent secondary but relevant areas.

6.4 Interpretation Guidelines

- Treat predictions as **guidance**, not definitive classification.
- Overlapping domains are common in interdisciplinary research.
- Slight variations in input (title/abstract wording) may lead to different predictions.

6.5 Limitations

- The model relies only on the provided text (title and abstract)
- It may not capture nuanced contributions or niche subfields
- Performance depends on the training data distribution

7 Common Issues

7.1 System Not Ready

Issue: The application is still initializing or loading models.

Solution:

- Wait until initialization is complete
- Ensure backend services (e.g., model server) are running

7.2 No Output

Issue: No predictions are displayed after execution.

Possible Causes:

- Empty or missing title/abstract
- Invalid input format

Solution:

- Ensure both fields are filled
- Avoid extremely short or non-informative text

7.3 CSV Errors (Bulk Mode)

Issue: File upload fails or processing does not start.

Checklist:

- File format is `.csv`
- Required columns are present:

  - `title`
  - `abstract`
- No missing or empty rows
- Proper encoding (UTF-8)

7.4 Incorrect or Irrelevant Predictions

Possible Causes:

- Vague or ambiguous abstract
- Missing technical keywords

Solution:

- Provide more detailed and specific descriptions
- Include methodology and domain-specific terms

7.5 Slow Processing

Cause:

- Large input size (long abstracts or large CSV files)
- High system load

Solution:

- Reduce input size where possible
- Split large datasets into smaller batches

8 Best Practices

8.1 Input Quality

- Provide clear, concise, and informative titles
- Include detailed abstracts describing:
  - Problem statement
  - Methodology
  - Key results or contributions

8.2 Data Formatting

- Ensure CSV files are clean and well-structured
- Use consistent column names (`title`, `abstract`)
- Remove duplicate or incomplete entries

8.3 Efficient Usage

- Start with small inputs to verify correctness
- Use bulk mode for large datasets
- Avoid interrupting processing once started